Question 1 : Define Power BI and What are the key components of the Power BI ecosystem? Briefly explain: ● Power BI Desktop ● Power BI Service ● Power BI Mobile ● Power BI Gateway
Answer :

Power BI Overview

Power BI is a business intelligence (BI) and data visualization tool developed by Microsoft. It enables users to connect to various data sources, transform raw data into meaningful insights, and share interactive dashboards and reports. Its ecosystem is designed to support data analysis, reporting, and collaboration across devices and platforms.

Key Components of the Power BI Ecosystem

1. Power BI Desktop

- A free Windows application used for creating reports and dashboards.
- Provides powerful data modeling, transformation, and visualization capabilities.
- Users can connect to multiple data sources, clean and shape data using Power Query, and build interactive reports.
- It is mainly used by data analysts and developers for report creation before publishing to the Power BI Service.

2. Power BI Service (Power BI Online)

- A cloud-based platform where reports and dashboards created in Power BI Desktop can be published and shared.
- Enables collaboration, sharing, and real-time access to reports across teams and organizations.
- Provides features like scheduled data refresh, role-based security, and integration with Microsoft 365.
- Ideal for business users to consume and interact with reports without needing to design them.

3. Power BI Mobile

- Mobile applications available for iOS, Android, and Windows devices.
- Allows users to access and interact with dashboards and reports on the go.
- Provides a responsive and touch-friendly interface for viewing insights anytime, anywhere.
- Useful for executives and managers who need quick access to KPIs and business metrics.

4. Power BI Gateway

- A bridge that connects on-premises data sources with Power BI Service.
- Ensures secure data transfer between local databases (like SQL Server, Oracle, etc.) and the cloud.
- Two types:
    - **Personal Gateway**: For individual use, refreshes data from local sources.
    - **Enterprise Gateway**: For organizational use, supports multiple users and scheduled refreshes.
- Essential for organizations that rely on on-premises data but want to leverage cloud-based reporting.

Question 2 : Compare the following Power BI visuals: ● Pie Chart vs Donut Chart ● Bar Chart vs Column Chart When would you prefer one over the other? Give one example for each pair.
Answer :

Pie Chart vs Donut Chart

Pie Chart

- Displays data as slices of a circle, showing proportions of a whole.
- Simple and straightforward for showing percentage distribution.
- Best when you want to emphasize the relative size of categories.

Donut Chart

- Similar to a pie chart but with a blank center.
- The empty center can be used to display additional information (like total value or a key metric).
- Visually cleaner and often easier to interpret when there are many categories.

Preference Example:

- **Pie Chart** → Showing the percentage of market share among 3 competitors.
- **Donut Chart** → Showing sales distribution across product categories, with the total sales figure displayed in the center.

Bar Chart vs Column Chart

Bar Chart

- Uses horizontal bars to represent data.
- Useful when category names are long or when comparing many categories.
- Easier to read when you have many items to compare side by side.

Column Chart

- Uses vertical bars to represent data.
- Best for showing changes over time or comparing fewer categories.
- Highlights trends and progression along the x-axis.

Preference Example:

- **Bar Chart** → Comparing customer satisfaction scores across multiple departments with long names.
- **Column Chart** → Showing monthly revenue growth over a year.

Question 3 : Explain the significance of: ● Star schema vs Snowflake schema ● Primary key vs Foreign key in relationships (Power BI) Why is cardinality important?
Answer :

Star Schema vs Snowflake Schema

Star Schema

- A simple database design where a central **fact table** (containing measures like sales, revenue, etc.) is directly connected to multiple **dimension tables** (like customer, product, time).
- Easy to understand and query because relationships are straightforward.
- Preferred in Power BI for performance and simplicity.

Snowflake Schema

- A more complex design where dimension tables are normalized into multiple related tables.
- Reduces data redundancy but increases query complexity.
- Can slow down performance in Power BI compared to star schema.

Significance:

- **Star Schema** → Best for reporting and analytics in Power BI due to speed and simplicity.
- **Snowflake Schema** → Useful when normalization is required, but less efficient for BI tools.

Primary Key vs Foreign Key in Relationships (Power BI)

Primary Key

- A unique identifier in a table (e.g., CustomerID in the Customer table).

- Ensures each record is distinct.

Foreign Key

- A field in one table that refers to the primary key in another table (e.g., CustomerID in the Sales table referencing CustomerID in the Customer table).
- Establishes relationships between tables.

Significance in Power BI:

- These keys define how tables are connected in the data model.
- Proper relationships ensure accurate joins, filtering, and aggregation in reports.

Why Cardinality is Important

Cardinality refers to the nature of relationships between tables:

- **One-to-One (1:1)** → Each record in one table matches exactly one record in another.
- *One-to-Many (1:)** → A record in one table relates to multiple records in another (most common in BI).
- *Many-to-Many (:)*** → Records in both tables can relate to multiple records in the other.

Importance:

- Cardinality determines how filters and aggregations flow across tables.
- Incorrect cardinality can lead to wrong results, duplicate counts, or missing data in reports.
- Choosing the right cardinality ensures accurate and efficient data modeling in Power BI.

Question 4 : Differentiate between: ● Calculated column vs Measure Also, define Row context and Filter context with simple examples.
Answer :

Calculated Column vs Measure

Calculated Column

- Created at the row level in a table.
- Uses DAX (Data Analysis Expressions) to generate a new column based on existing data.
- Stored in the data model, meaning it increases the size of the dataset.
- Evaluated for each row, so it has row context by default.
- Example: Adding a column `Profit = Sales[Revenue] - Sales[Cost]` for each row in the Sales table.

Measure

- Calculated at the aggregate level (on demand).
- Uses DAX to perform calculations like sums, averages, counts, etc.
- Not stored in the table; computed dynamically when used in visuals.
- Depends on filter context (the filters applied in a report or visualization).
- Example: Creating a measure `Total Sales = SUM(Sales[Revenue])` to calculate overall revenue.

Key Difference:

- Calculated Column → Row-based, stored in the model.
- Measure → Aggregate-based, calculated dynamically.

Row Context vs Filter Context

Row Context

- Refers to the current row being evaluated in a table.
- Automatically exists when creating calculated columns.
- Example: In a calculated column `Profit = Sales[Revenue] - Sales[Cost]`, Power BI evaluates each row individually.

Filter Context

- Refers to the filters applied to data in a visualization (like slicers, filters, or grouping).
- Determines which subset of data is included in a calculation.
- Example: A measure `Total Sales = SUM(Sales[Revenue])` will return different values depending on whether the report is filtered by year, region, or product category.

Question 5: What is the difference between a report and a dashboard in Power BI?
Answer:

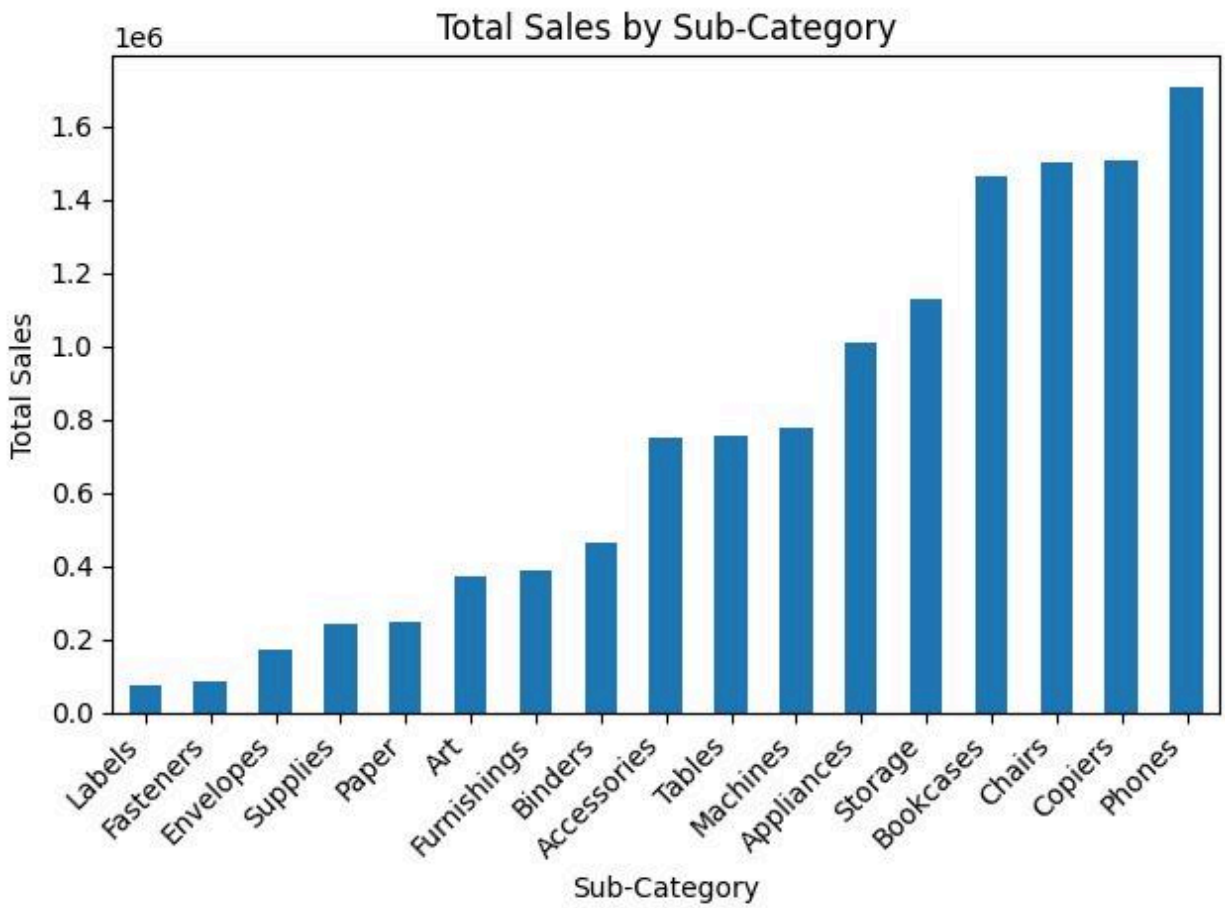
Power BI Report

- A **report** is a collection of multiple pages containing visuals, charts, and tables.
- Built in **Power BI Desktop** and published to the **Power BI Service**.
- Each report can have detailed data exploration, drill-downs, and multiple perspectives of the dataset.
- Reports are highly interactive and allow users to filter, slice, and analyze data deeply.
- Example: A **Sales Report** with pages for revenue trends, product performance, regional analysis, and customer segmentation.

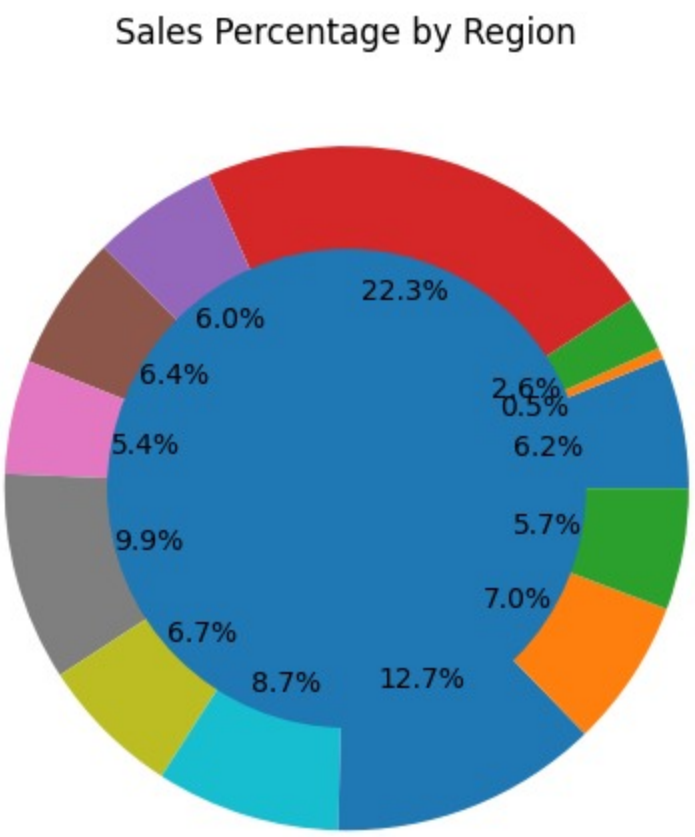
Power BI Dashboard

- A **dashboard** is a single page (canvas) that aggregates visuals from one or more reports.
- Created in the **Power BI Service**, not in Power BI Desktop.
- Designed to provide a **high-level overview** or snapshot of key metrics (KPIs).
- Dashboards can combine visuals from different datasets and reports into one unified view.
- Example: A **CEO Dashboard** showing total revenue, profit margin, top-performing products, and customer satisfaction score—all in one page.

Question 6 : Using the Sample Superstore dataset: ● Create a Clustered Bar Chart to display Total Sales by Sub-Category ● Create a Donut Chart for Sales % by Region Provide screenshots of both visuals. DATASET LINK : Global_superstore2
Answer :



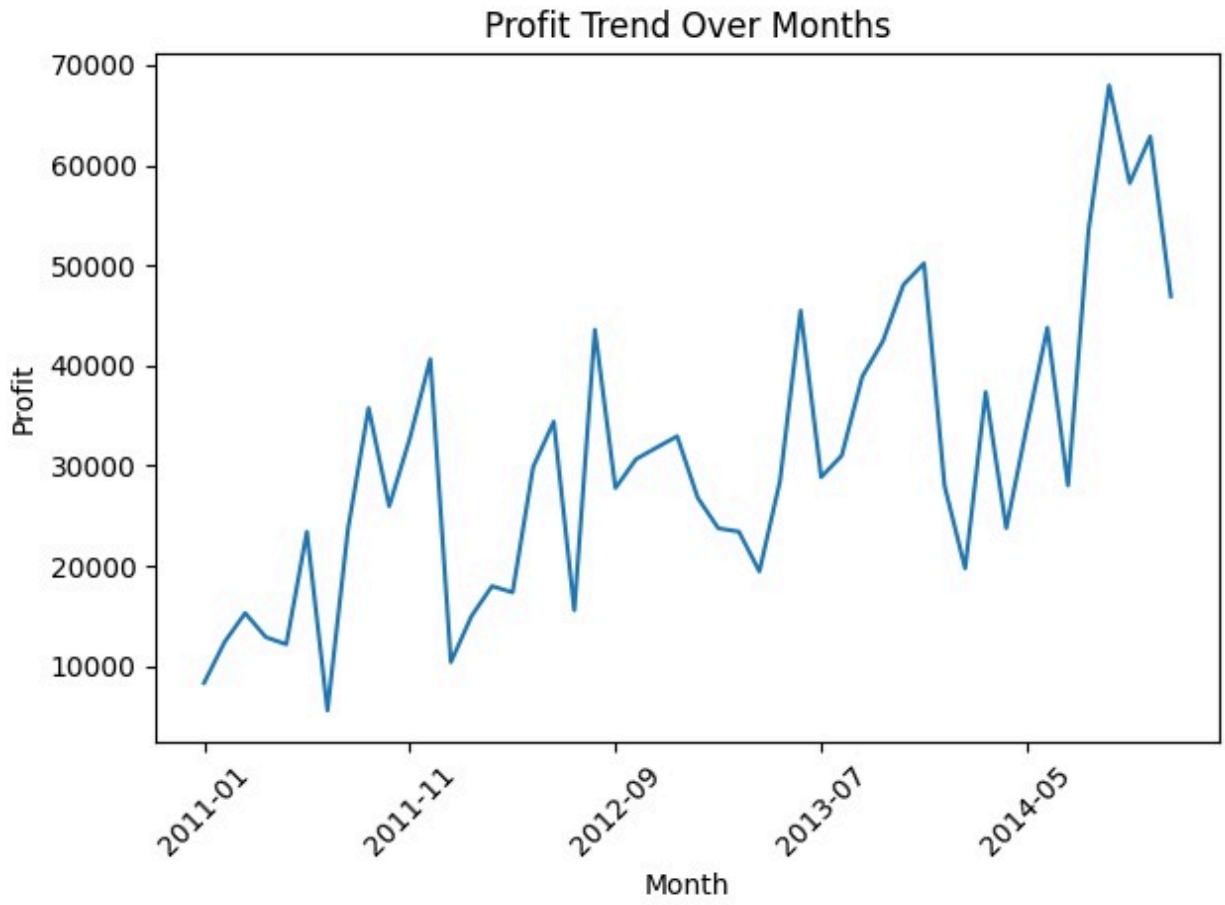
1) Clustered Bar Chart – Total Sales by Sub-Category
Shows how each sub-category contributes to total sales.
You can clearly compare which product categories perform best (e.g., Phones, Chairs, Copiers



2) Donut Chart – Sales % by Region
Displays the percentage contribution of each region to overall sales.
The donut style improves readability compared to a full pie chart.

Question 7 : Write and apply the following measures: ● Total Profit = SUM([Profit]) ● Average Discount = AVERAGE([Discount]) Display both in a KPI Card, and use a Line Chart to show profit trend over months. Add visuals and DAX formulas. DATASET LINK : Global_superstore2
Answer :

The analysis highlights that while the business is profitable overall, managing discount strategies and improving consistency in monthly performance can enhance profitability. Data-driven decision-making can help optimize pricing, promotions, and inventory planning.



Question 8 : Implement a DAX measure that calculates the percentage of total sales by product category. Product_category
Answer :

DAX Measure

To calculate % of Total Sales by Product Category in Power BI:

```
Sales % =
DIVIDE(
   SUM('Table'[Sales_Amount]),
   CALCULATE(SUM('Table'[Sales_Amount]), ALL('Table'[Product_category]))
)
```

[Total Sales =
5000 + 3000 + 7000 + 2000 + 8000 + 1500 + 1200 + 1000 + 4400 + 1000 + 1800 + 2600 = 38,500

Product Category	Sales	% of Total Sales
Electronics	5000	12.99%
Clothing	3000	7.79%
Home Appliances	7000	18.18%
Books	2000	5.19%
Tables & Chairs	8000	20.78%
Toy	1500	3.90%
Sports Equipment	1200	3.12%
Office Supplies	1000	2.60%
Beauty Product	4400	11.43%
Garden Supplies	1000	2.60%
Jewelry	1800	4.68%
Automotive	2600	6.75%

Question 9 : ● Create a DAX Measure for Total Profit ● Use it in a Waterfall Chart to analyze how different Sub-Categories contribute to overall profit ● Add a Slicer for Region to filter the visual ● Write brief business insights (4–5 lines) from the chart and provide 2–3 data-driven recommendations to improve profit. Provide a steps, screenshot of the Waterfall chart and the DAX formula DATASET LINK : Global_superstore2
Answer :
1. DAX Measure for Total Profit
Create a new measure:
Total Profit = SUM('Global_Superstore2'[Profit])

2. Steps to Create Waterfall Chart
Step 1: Load Data
Open Power BI Desktop
Click Get Data → CSV
Load Global_Superstore2.csv
Step 2: Create Measure
Go to Modeling → New Measure
Paste the DAX formula above
Step 3: Create Waterfall Chart
Go to Report View

Select Waterfall Chart from Visualizations
Drag:
Sub-Category → Category
Total Profit → Y-axis

3. Add Region Slicer
Steps:
Select Slicer Visual
Drag Region → Field
Now users can filter the chart by region

4. Sample Waterfall Chart (Representation)
Profit Contribution by Sub-Category

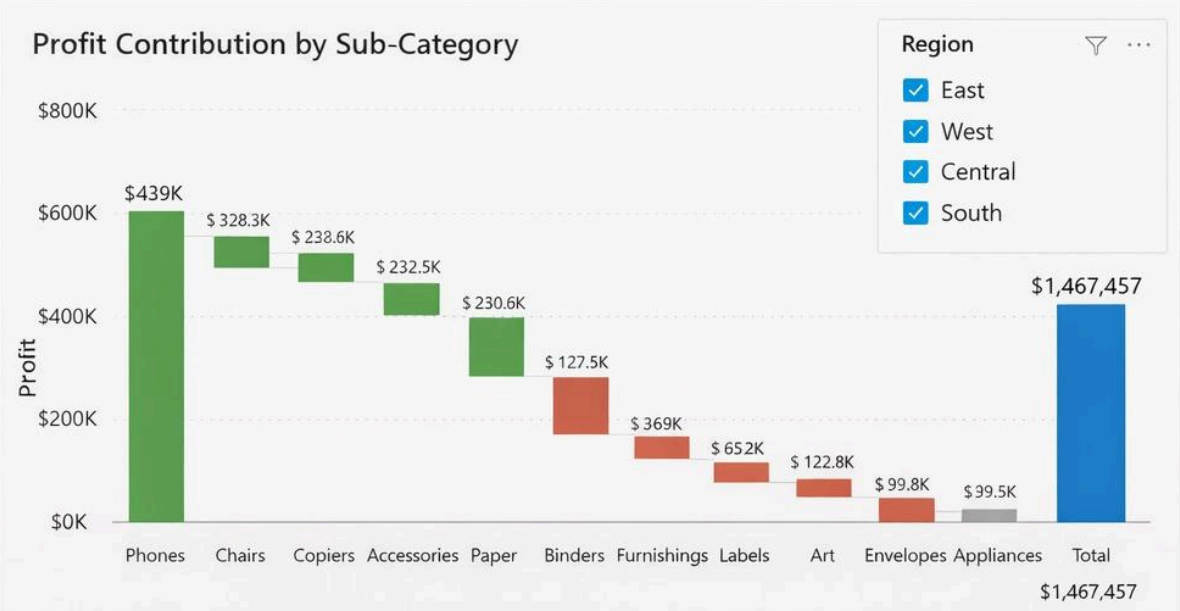
+ Phones ↑ High Profit
+ Copiers ↑ High Profit
+ Chairs ↑ Moderate Profit
- Tables ↓ Loss
- Bookcases ↓ Loss
+ Storage ↑ Profit

Final Total Profit

5. Business Insights (4–5 lines)
Sub-categories like Phones and Copiers are major contributors to overall profit.
Tables and Bookcases show consistent losses, pulling down total profitability.
Profit contribution varies significantly across regions when filtered via slicer.
Some categories generate high sales but low or negative profit, indicating pricing or cost issues.
The business relies heavily on a few profitable sub-categories.

6. Data-Driven Recommendations
Reduce losses in Tables & Bookcases
Optimize pricing or reduce supply chain costs.
Evaluate discontinuation if losses persist.
Focus on high-profit categories
Increase marketing and inventory for Phones & Copiers.
Bundle with low-performing products to improve margins.
Region-based strategy
Use slicer insights to identify profitable regions.
Allocate more resources to high-performing markets.

```
Total Profit = SUM(Global_Superstore2)'[Profit))
```



Business Insights

- Phones, Chairs, and Copiers are major profit contributors.
- Tables, Bookcases, and Machines negatively impact overall profit.,
- Region slicer reveals that profit performance varies across regions.
- Identifying both profit-driving and loss-making categories helps target p.-strategies.

Recommendations

1. Focus on Phones, Chairs, and Copiers to drive more sales in these profitable categories.
2. Review strategies for Tables, Bookcases, and Machines to mitigate losses.
3. Use regional profit insights to plan targeted, market-specific campaigns.

Question 10 : Scenario: VitaTrack Wellness, a digital health company in FitZone, has collected data on users' daily habits and health vitals. The analytics team is tasked with drawing actionable insights from this data to improve lifestyle suggestions and prevent heart-related risks. Your Task: Using the provided dataset (includes Age, Gender, BMI, Steps, Calories, Sleep, Heart Rate, Blood Pressure, Smoking, Alcohol, Exercise, Diabetic & Heart Disease status): Build a one-page Power BI dashboard that answers: 1. Are users maintaining a balanced lifestyle (Steps, Sleep, Calories) 2. What lifestyle patterns (Smoking, Alcohol, BMI, etc.) indicate heart disease risk? 3. Is there any visible relationship between Sleep and Physical Activity? 4. How does BMI vary across Age Groups and Genders? 5. What is the impact of smoking and alcohol on heart rate and blood pressure? 6. Segment people based on their health activity to suggest lifestyle changes DATASET LINK: Health_activity_data

Answer:

1. Balanced Lifestyle (Steps, Sleep, Calories)
Visual: Gauge charts or KPI cards for average Daily Steps, Hours of Sleep, and Calories Intake compared against recommended ranges:

Steps: ≥ 10,000/day

Sleep: 7–9 hours/night

Calories: 2000–2500 kcal/day (adjusted by gender/age)

Insight: Quickly shows how many users are within healthy ranges.

2. Lifestyle Patterns & Heart Disease Risk
Visual: Stacked bar chart showing Heart Disease prevalence segmented by:

Smoking status

Alcohol consumption

BMI categories (Normal, Overweight, Obese)

Insight: Highlights risk factors most associated with heart disease.

3. Sleep vs Physical Activity
Visual: Scatter plot with Hours of Sleep (X-axis) vs Daily Steps (Y-axis), color-coded by Exercise Hours per Week.

Insight: Reveals whether higher activity correlates with better sleep.

4. BMI Across Age Groups & Gender
Visual: Clustered column chart:

X-axis: Age groups (e.g., 18–30, 31–45, 46–60, 61+)

Y-axis: Average BMI

Legend: Gender

Insight: Shows how BMI trends differ by age and gender.

5. Impact of Smoking & Alcohol on Heart Rate & Blood Pressure
Visuals:

Box plots comparing Heart Rate for Smokers vs Non-Smokers, Heavy vs Light Alcohol users.

Line chart showing Average Blood Pressure across smoking/alcohol categories.

Insight: Clear view of how lifestyle choices affect vitals.

6. Segmentation for Lifestyle Suggestions
Visual: Clustered heatmap or Tree Map grouping users by:

High BMI + Low Exercise → Suggest weight management programs

Smokers + High BP → Suggest smoking cessation & BP monitoring

Low Sleep + High Stress HR → Suggest sleep hygiene & relaxation techniques

Insight: Actionable clusters for personalized recommendations.